

# ROBOT PROGRAMMING FRAMEWORKS AND IOT PLATFORMS

**Scuola Superiore Sant'Anna**

Anno Accademico 2025/2026

Gastone Ciuti



# Organization

|            |                                                                                                                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11/11/2025 | Introduction and Communication Protocols (1/2) (3 hours – SLOT 1)                                                                                                                                                        |
| 17/11/2025 | Communication Protocols (2/2) (3 hours – SLOT 1)                                                                                                                                                                         |
| 18/11/2025 | IoT Platforms and Main Features (3 hours – SLOT 2)                                                                                                                                                                       |
| 24/11/2025 | Set-up of Particle Development Boards & Basic Functions (2 hours) <ul style="list-style-type: none"><li>• Web-based programming, cloud functions and examples</li></ul>                                                  |
| 25/11/2025 | How to Program an IoT System: Functions and Coding (1/3) (3 hours – SLOT 3) <ul style="list-style-type: none"><li>• Other Programming Environments, <i>i.e.</i> Particle Workbench based on Visual Studio Code</li></ul> |
| 01/12/2025 | How to Program an IoT System: Functions and Coding (2/3) (2 hours – SLOT 3) <ul style="list-style-type: none"><li>• Advanced Programming Functionalities for IoT Platforms and Use Cases</li></ul>                       |
| 02/12/2025 | How to Program an IoT System: Functions and Coding (3/3) (3 hours – SLOT 3) <ul style="list-style-type: none"><li>• Integration with Web Services, <i>i.e.</i> IFTTT (if this than that)</li></ul>                       |
| 09/12/2025 | Examples with IoT Systems and Exercises (3 hours – SLOT 4)                                                                                                                                                               |



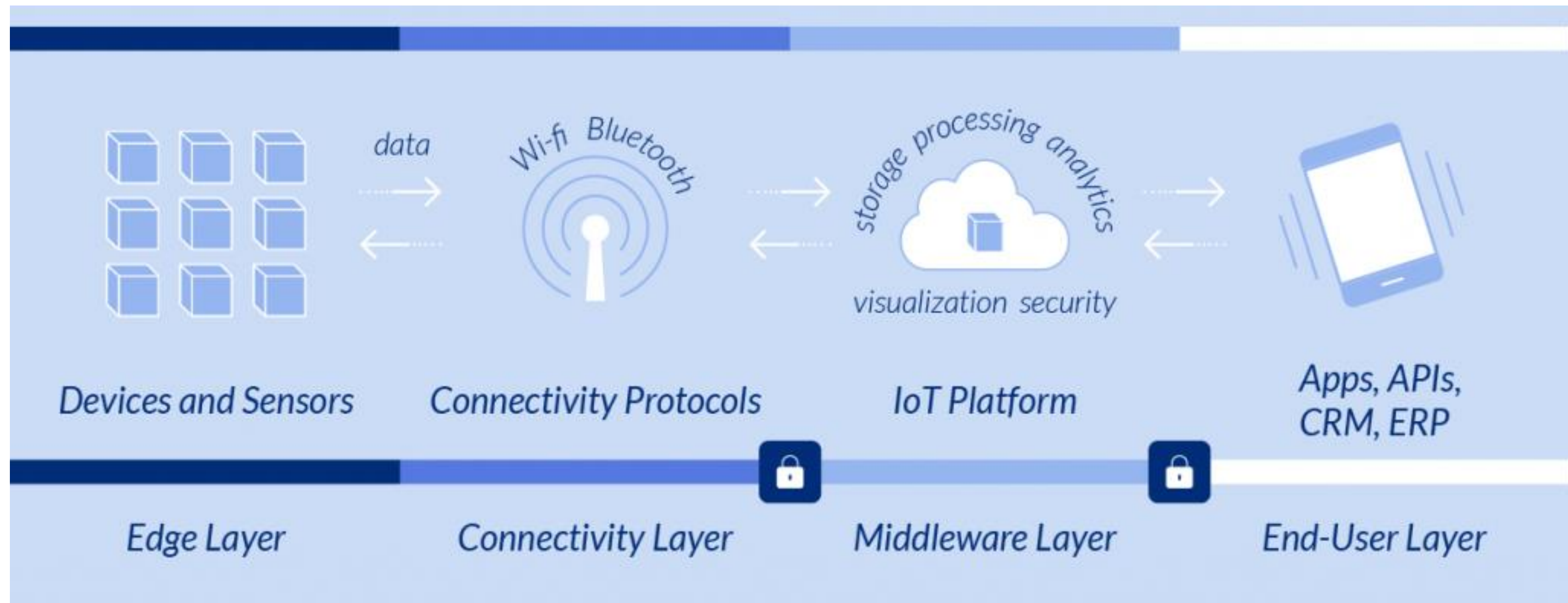
# IoT Platforms and Main Features (24.11.2025) – SLOT 2

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# What is an IoT device?

**Internet of things (IoT):** “An open and comprehensive network of intelligent objects that have the capacity to auto-organize, share information, data and resources, reacting and acting in face of situations and changes in the environment”.

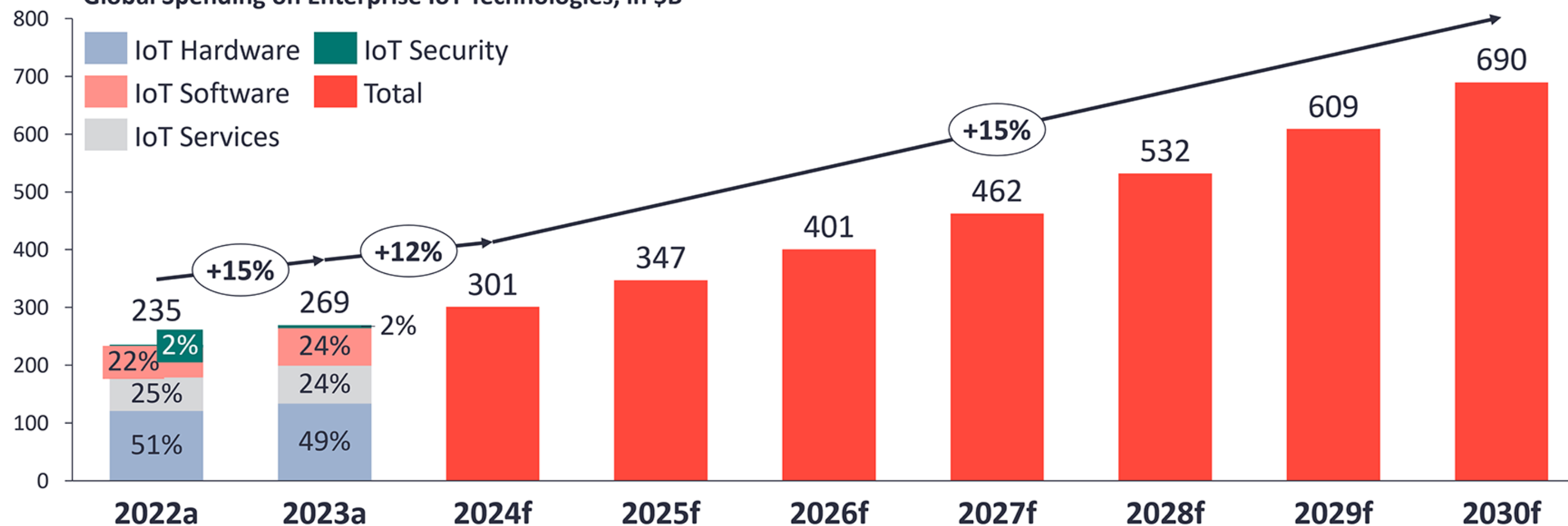


# Internet of Things Uses By Industry



# The enterprise IoT market by technology 2023–2030

Data as of June 2024

**Global Spending on Enterprise IoT Technologies, in \$B**


**Note:** IoT Analytics defines IoT as a network of internet-enabled physical objects. Objects that become internet-enabled (IoT devices) typically interact via embedded systems, some form of network communication, or a combination of edge and cloud computing.

The data from IoT-connected devices is often used to create novel end-user applications. Connected personal computers, tablets, and smartphones are not considered IoT, although these may be part of the solution setup.

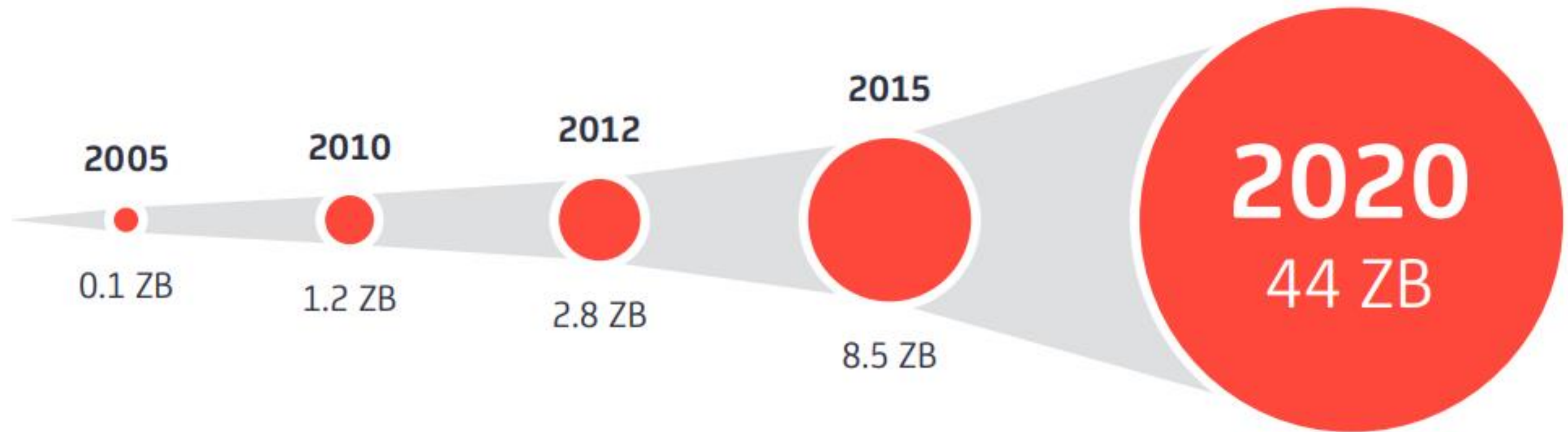
Devices connected via extremely simple connectivity methods, such as radio frequency identification or quick response codes, are not considered IoT devices. Since the last update in 2023 our definition of the enterprise IoT tech stack slightly changed.

a: Actuals, f: Forecast

**Source:** IoT Analytics Research 2024 – Global IoT Enterprise Spending Dashboard (Q2/2024 update). We welcome republishing of images but ask for source citation with a link to the original post or company website.



# 100 zettabytes by 2025



**EXHIBIT 2:** Explosion of data in the world – Data created by people, enterprises, and things will attain 44 zettabytes by 2020 (Source: Adapted from IDC<sup>7</sup>)

**1 ZB is equal to  $\sim 1 \times 10^9$  Terabyte (TB)**



# Connectivity protocols for IoT systems (examples)



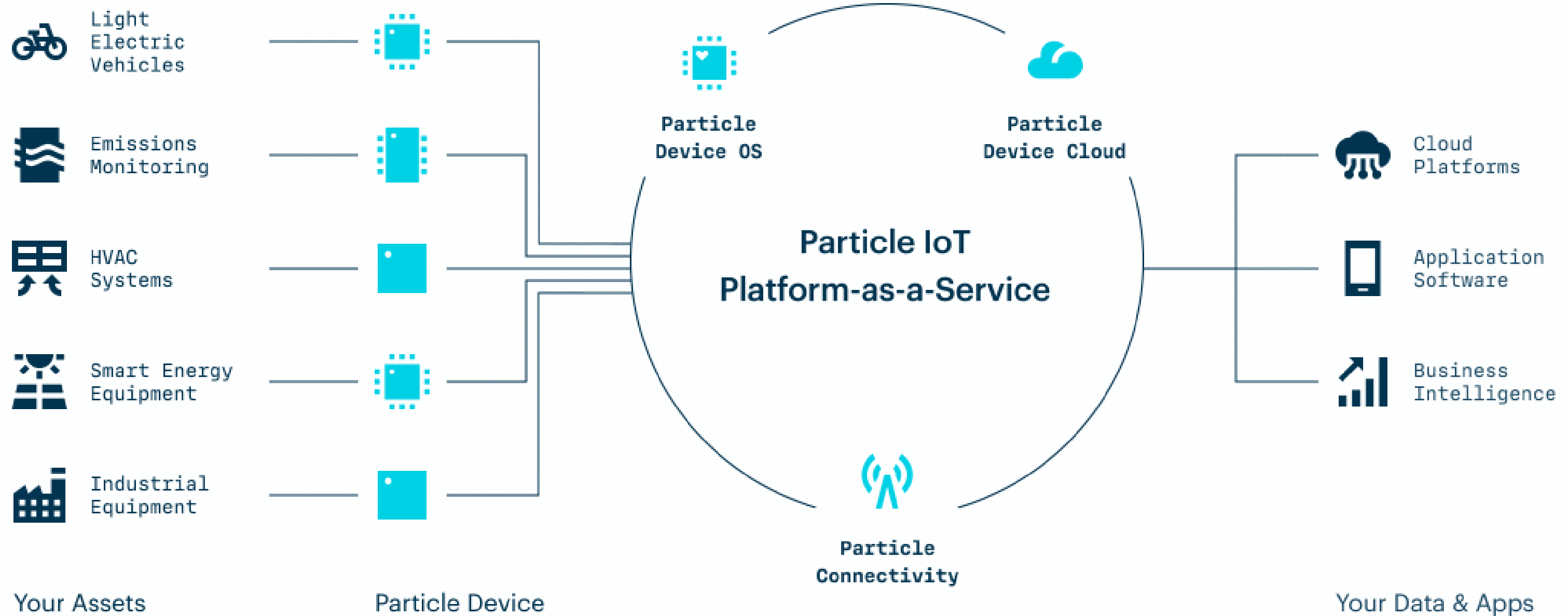
**WiFi-BLE**



**Multi-Radio**



# Particle: a complete edge-to-cloud platform



Trusted by 8500+ industry titans and disruptors



# An example of IoT device: Particle ARGON (HW specs)



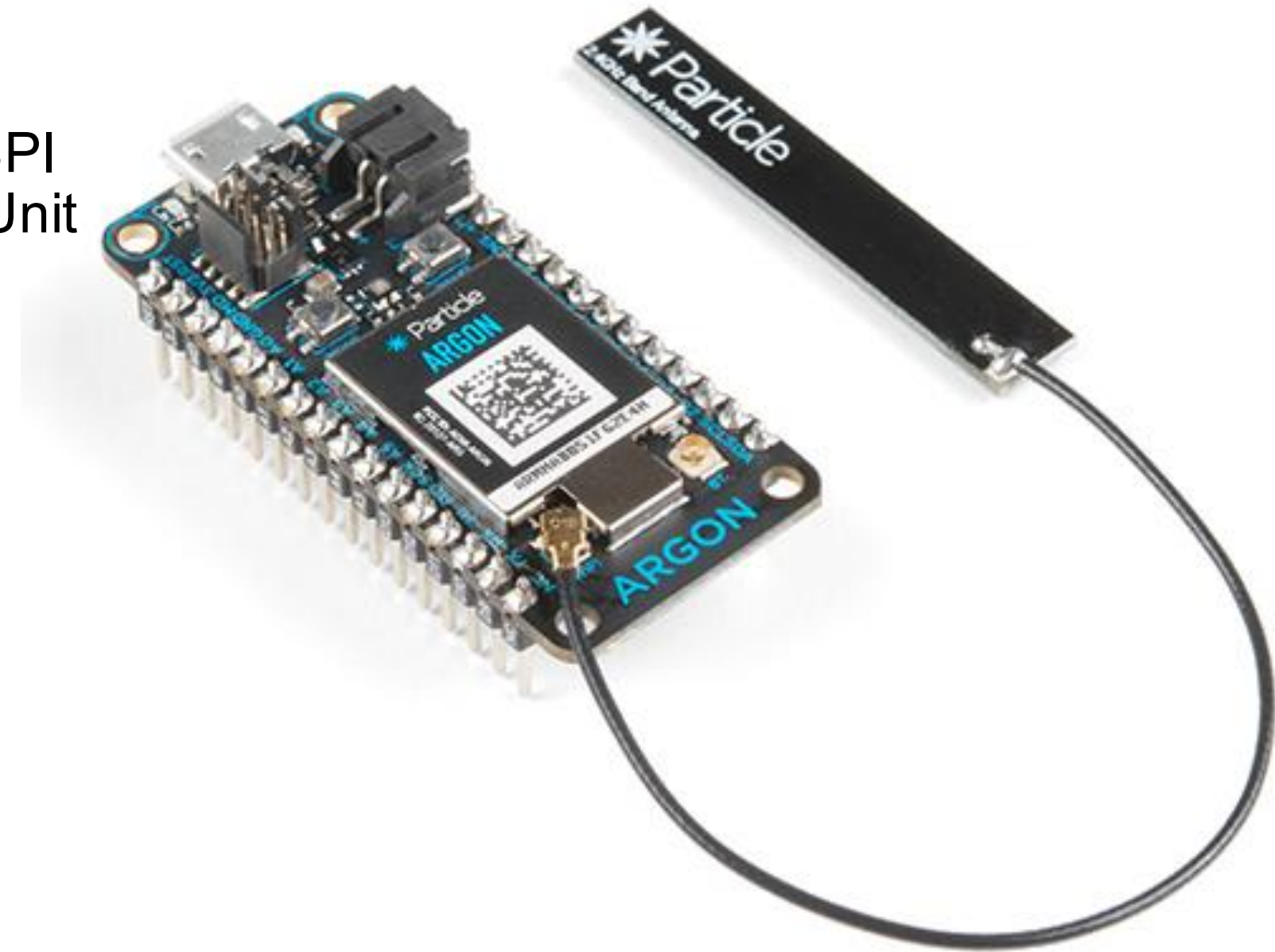
## Nordic Semiconductor nRF52840 SoC

- ARM Cortex-M4F 32-bit processor @ 64MHz
- 1MB flash, 256KB RAM
- Bluetooth LE (BLE) central and peripheral support
- 20 mixed signal GPIO (6 x Analog, 8 x PWM), UART, I2C, SPI
- Supports DSP instructions, HW accelerated Floating Point Unit (FPU) calculations
- ARM TrustZone CryptoCell-310 Cryptographic and security module
- Up to +8 dBm TX power (down to -20 dBm in 4 dB steps)
- NFC-A radio

## Argon Wi-Fi network coprocessor

### Espressif ESP32-D0WD 2.4 GHz Wi-Fi coprocessor

- On-board 4MB flash for the ESP32
- 802.11 b/g/n support
- 802.11 n (2.4 GHz), up to 150 Mbps

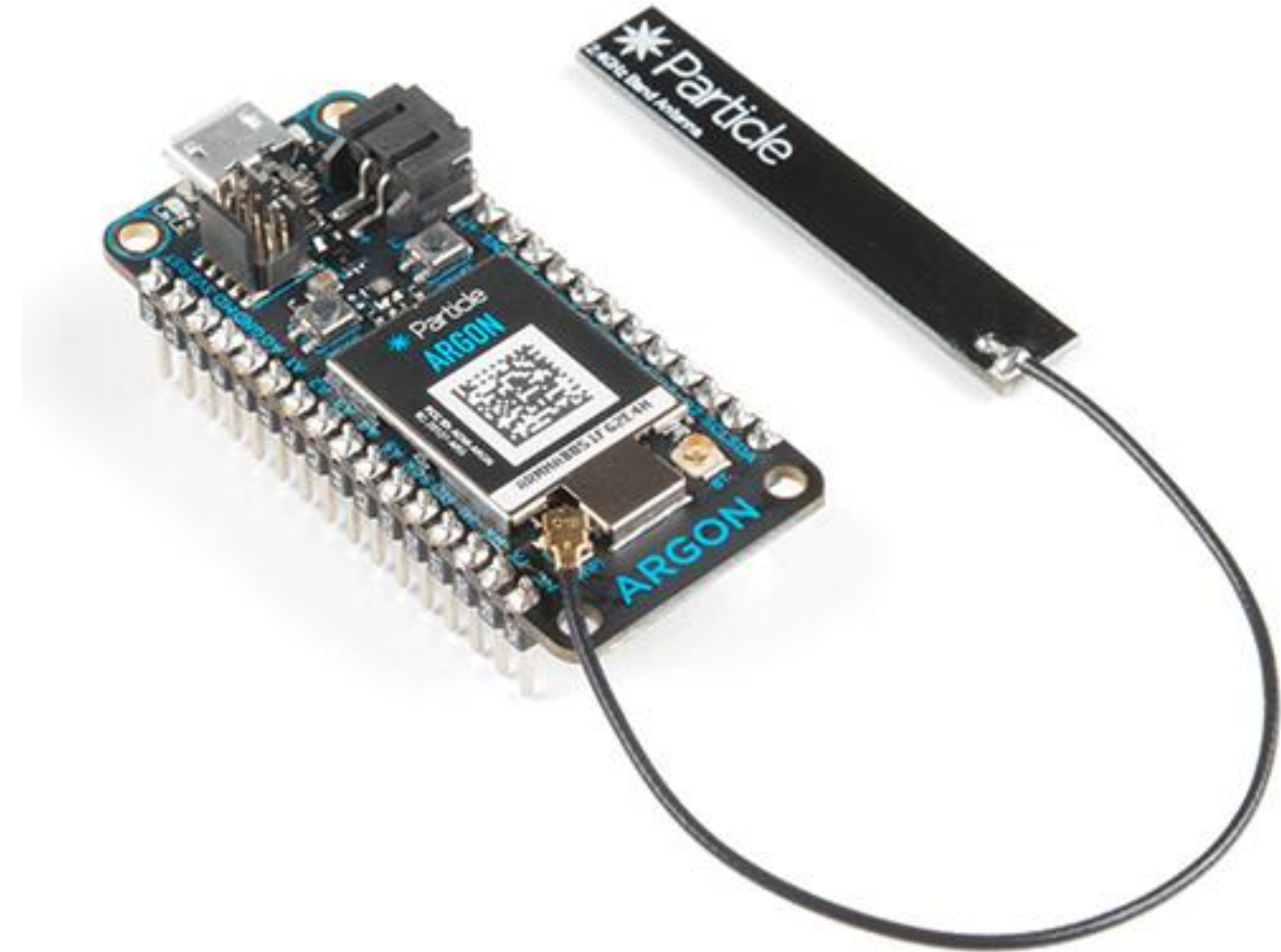


# An example of IoT device: Particle ARGON (HW specs)

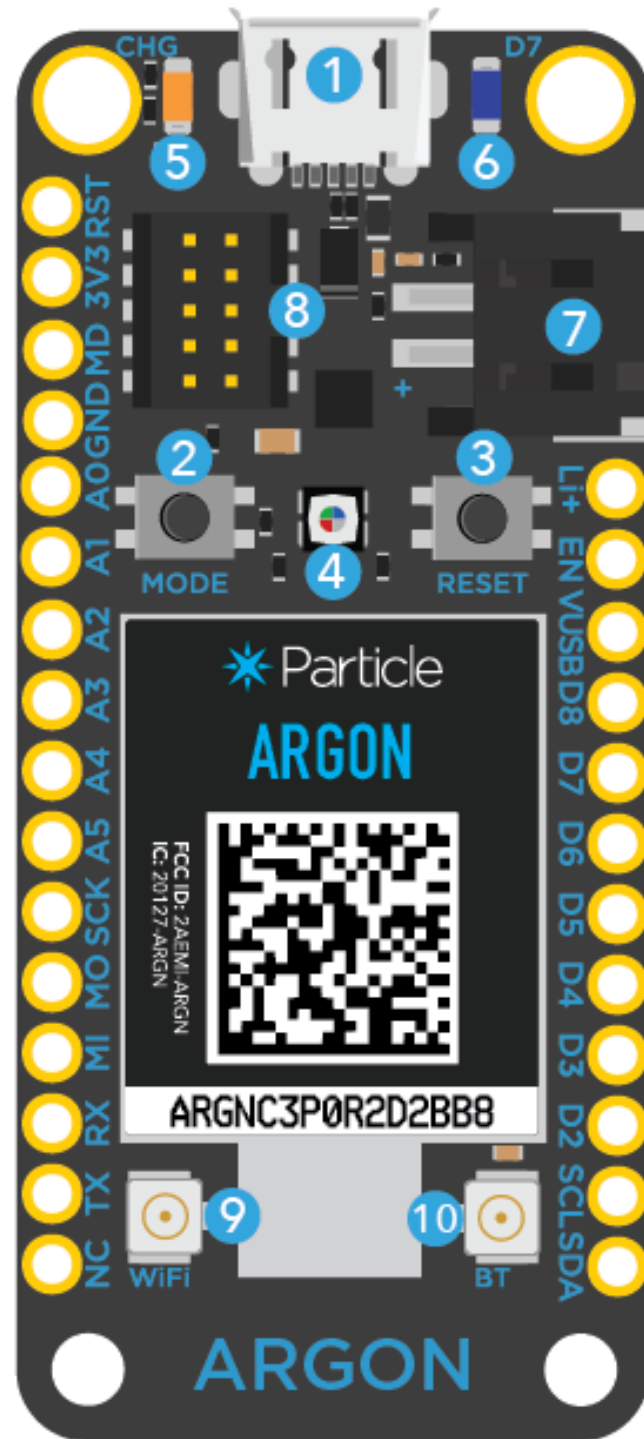


## Argon general specifications:

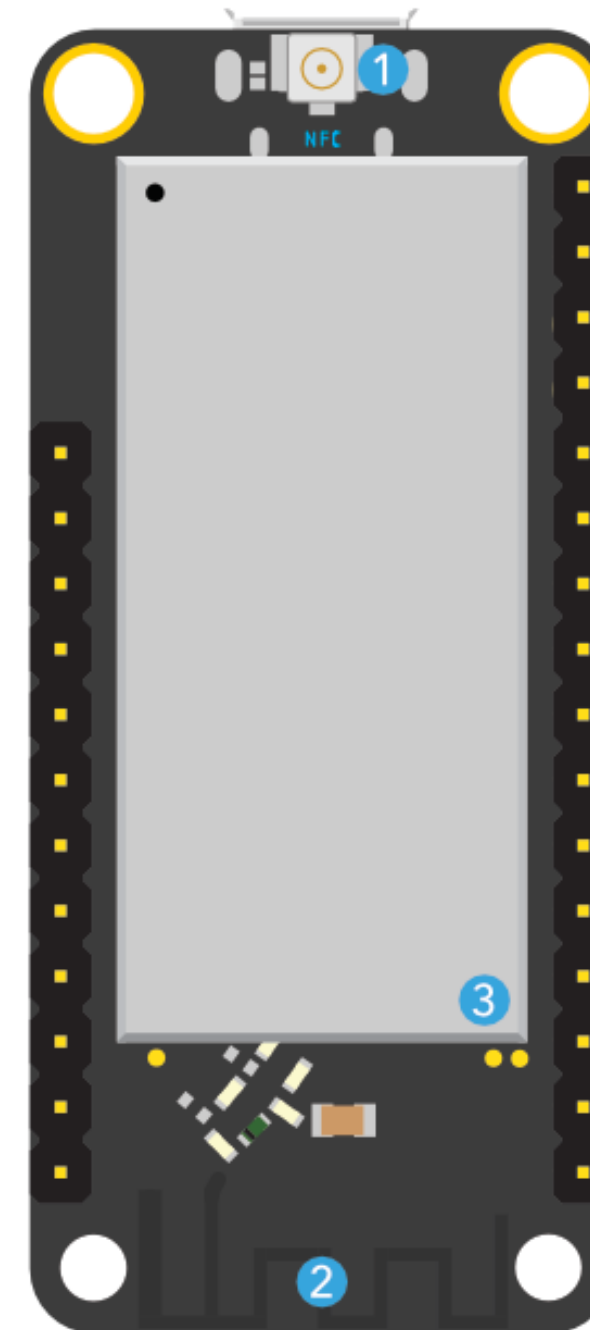
- On-board additional 4MB SPI flash
- Micro USB 2.0 full speed (12 Mbps)
- Integrated Li-Po charging and battery connector
- JTAG (SWD) Connector
- RGB status LED
- Reset and Mode buttons
- On-board 2.4GHz PCB antenna for Bluetooth (does not support Wi-Fi)
- Two U.FL connectors for external antennas (one for Bluetooth, another for Wi-Fi)
- Meets the specification in dimensions and pinout (<https://learn.adafruit.com/adafruit-feather/feather-specification>)
- FCC, CE and IC certified
- RoHS compliant (lead-free)



# An example of IoT device: Particle ARGON (HW specs)



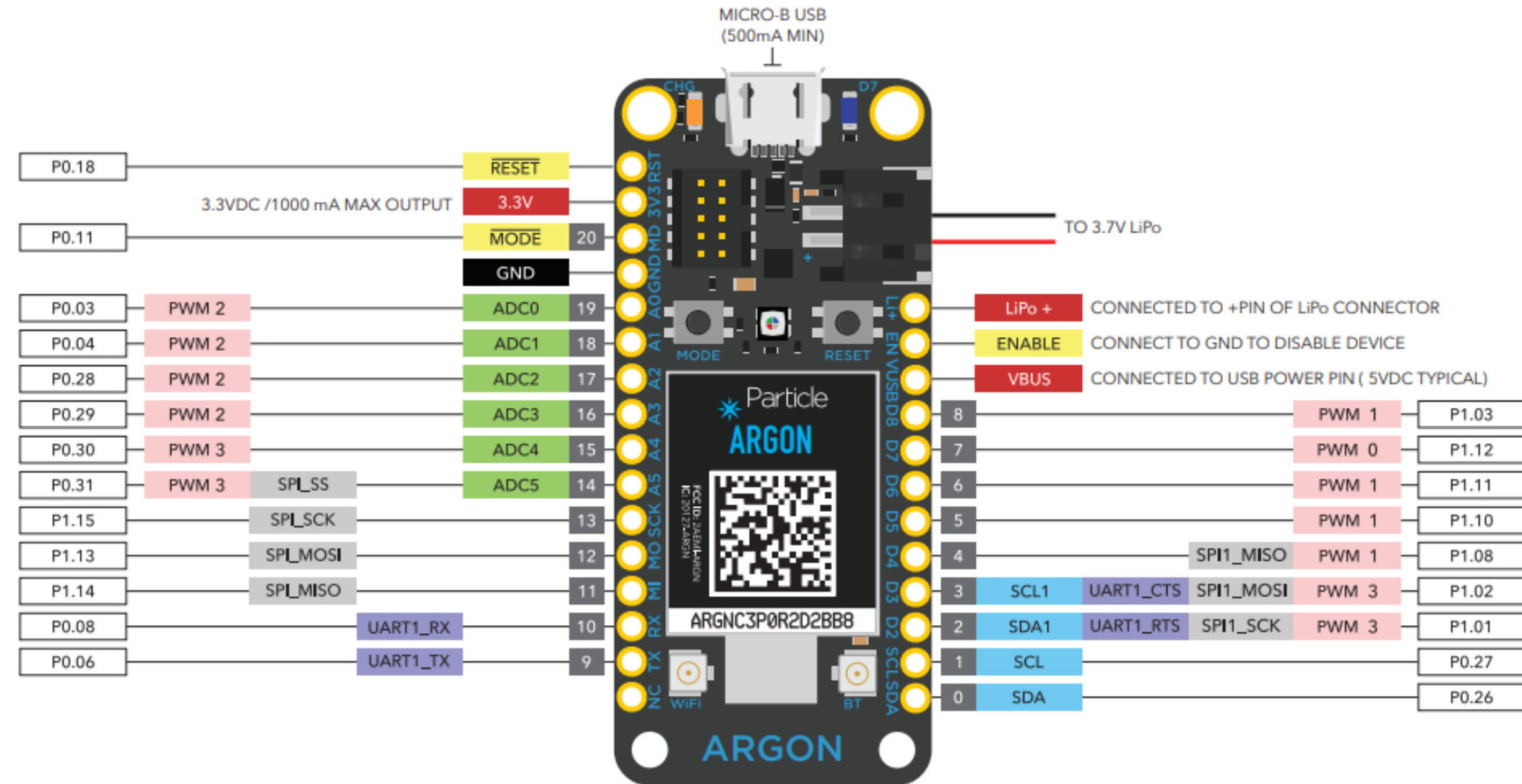
- 1 USB
- 2 MODE BUTTON
- 3 RESET BUTTON
- 4 RGB STATUS LED
- 5 CHARGE STATUS LED
- 6 USER LED (ON D7)
- 7 LiPo BATT CONNECTOR
- 8 DEBUG CONNECTOR (SWD)
- 9 u.FL CONNECTOR FOR WiFi ANTENNA (REQUIRED)
- 10 u.FL CONNECTOR FOR EXTERN/ MESH ANTENNA (OPTIONAL)



- 1 u.FL CONNECTOR FOR NFC ANTENNA
- 2 ON-BOARD PCB ANTENNA FOR MESH
- 3 RF SHIELD

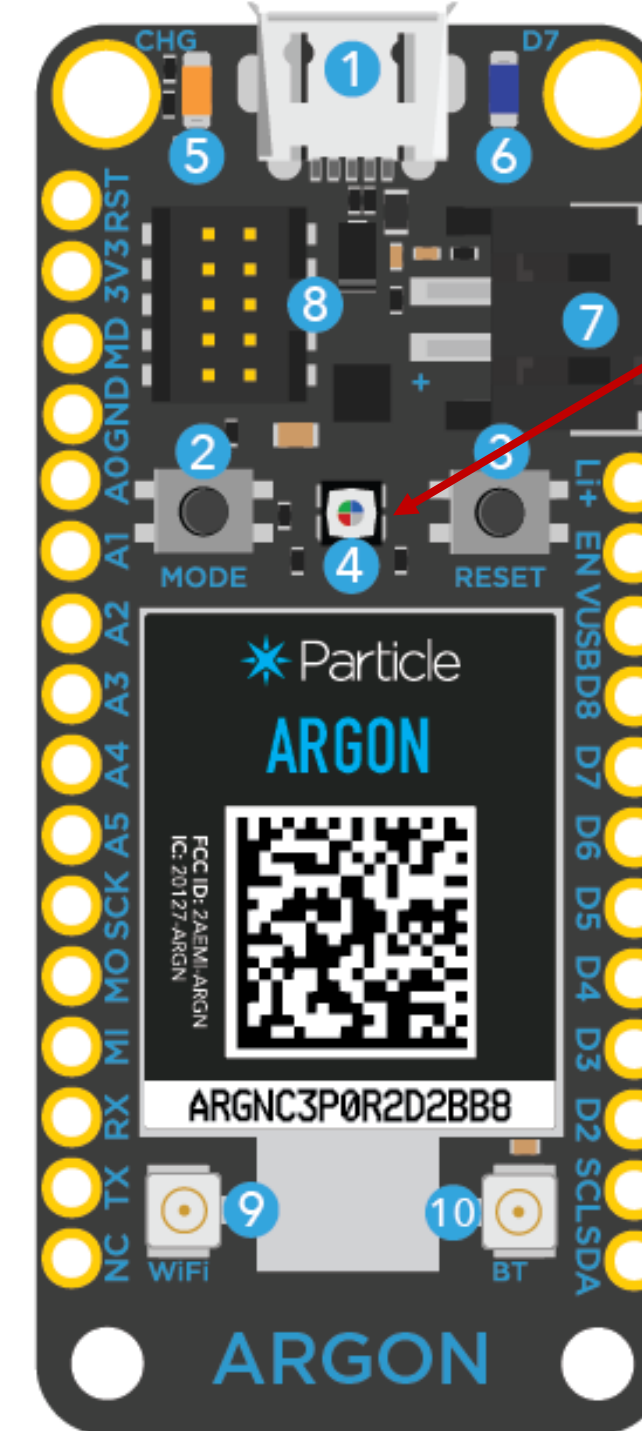


# Pinout diagram



# Pin description and LED status

| Pin       | Description                                                                                 |
|-----------|---------------------------------------------------------------------------------------------|
| Li+       | This pin is internally connected to the positive terminal of the LiPo battery connector.    |
| VUSB      | This pin is internally connected to the USB (+ve) supply.                                   |
| 3V3       | This pin is the output of the on-board 3.3V regulator.                                      |
| GND       | System ground pin.                                                                          |
| EN        | Device enable pin is internally pulled-up. To disable the device, connect this pin to GND.  |
| RST       | Active-low system reset input. This pin is internally pulled-up.                            |
| MD        | This pin is internally connected to the MODE button. The MODE function is active-low.       |
| RX        | Primarily used as UART RX, but can also be used as a digital GPIO.                          |
| TX        | Primarily used as UART TX, but can also be used as a digital GPIO.                          |
| SDA       | Primarily used as data pin for I2C, but can also be used as a digital GPIO.                 |
| SCL       | Primarily used as clock pin for I2C, but can also be used as a digital GPIO.                |
| MO,MI,SCK | These are the SPI interface pins, but can also be used as a digital GPIO.                   |
| D2-D8     | These are generic GPIO pins. D2-D8 are PWM-able.                                            |
| A0-A5     | These are analog input pins that can also act as standard digital GPIO. A0-A5 are PWM-able. |



LED status

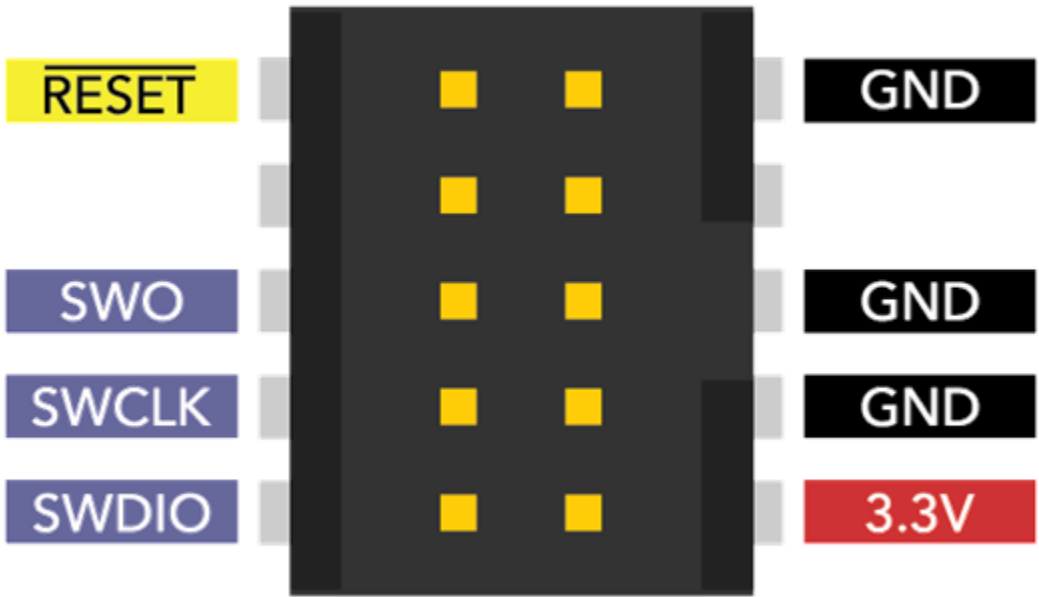


# Peripherals, GPIO and SWD

| Peripheral Type | Qty | Input(I) / Output(O) |
|-----------------|-----|----------------------|
| Digital         | 20  | I/O                  |
| Analog (ADC)    | 6   | I                    |
| UART            | 1   | I/O                  |
| SPI             | 1   | I/O                  |
| I2C             | 2   | I/O                  |
| USB             | 1   | I/O                  |
| PWM             | 8   | O                    |

All GPIOs are only rated at 3.3VDC max.

The Argon has a dedicated 10 pin debug connector that exposes the SWD interface of the nRF5280. This interface can be used to **debug your code or reprogram your Argon bootloader, device OS, or the user firmware** using any standard SWD tools as [LINK](#).

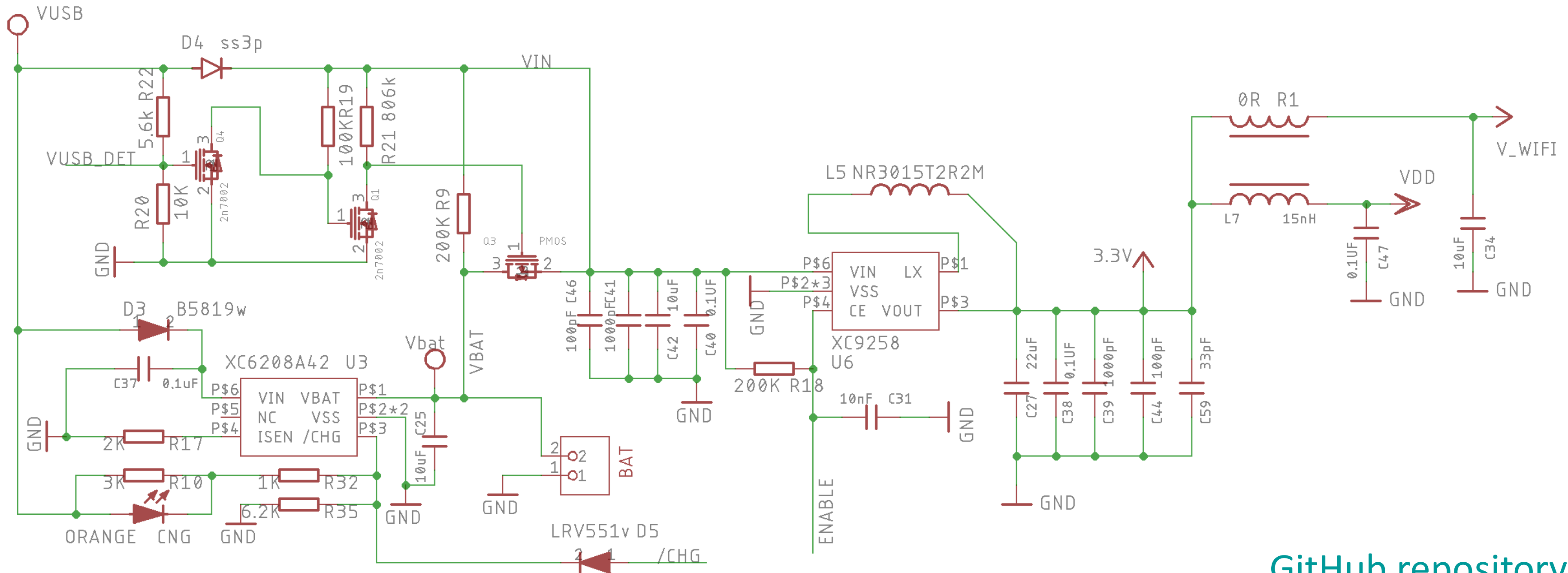


Serial Wire Debug (SWD)

| Pin   | Type              | Explanation                                                                                                                                                                                                                                       |
|-------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SWCLK | Input             | Clock signal to target CPU. It is recommended that this pin is pulled to a defined state of the target board. Shared with TMS on targets which support JTAG and SWD.                                                                              |
| SWDIO | I/O               | Bi-directional data pin. This pin should be pulled up on the target. Shared with TMS on targets which support JTAG and SWD.                                                                                                                       |
| SWO   | Output (optional) | Serial wire output, allowing the CPU to output custom data like printf output, asynchronous to SWCLK and other debug pins. Usually shared with the TDO pin on devices which support JTAG and SWD. Not available on all devices which support SWD. |



# Electronic Schemes



[GitHub repository](#)



# Technical specifications

## Absolute maximum ratings

| Parameter                                | Symbol                 | Min | Typ | Max  | Unit |
|------------------------------------------|------------------------|-----|-----|------|------|
| Supply Input Voltage                     | V <sub>IN-MAX</sub>    |     |     | +6.2 | V    |
| Battery Input Voltage                    | V <sub>LiPo</sub>      |     |     | +6.5 | V    |
| Supply Output Current                    | I <sub>3V3-MAX-L</sub> |     |     | 1000 | mA   |
| Storage Temperature                      | T <sub>stg</sub>       | -30 |     | +75  | °C   |
| ESD Susceptibility HBM (Human Body Mode) | V <sub>ESD</sub>       |     |     | 2    | kV   |

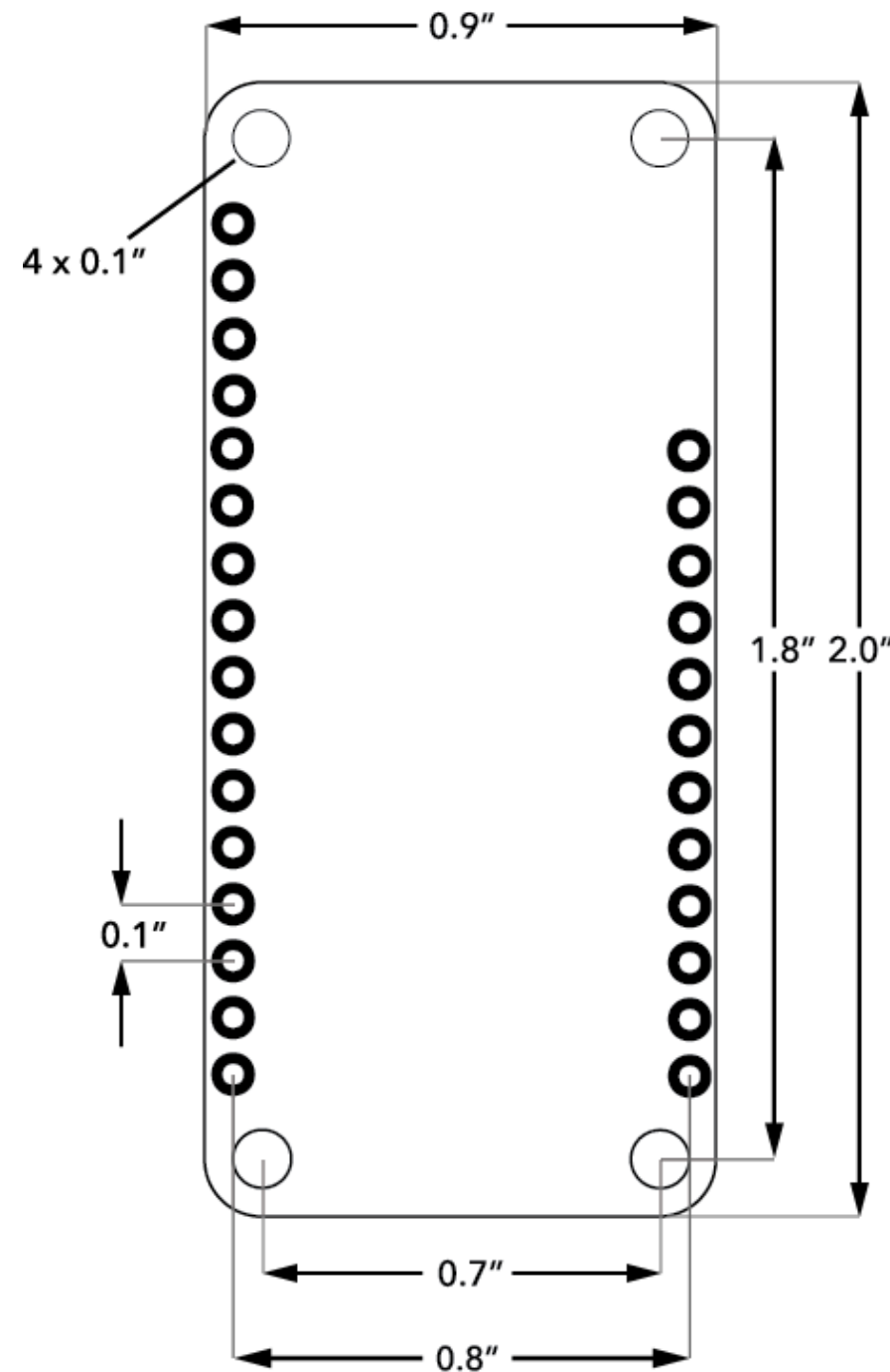
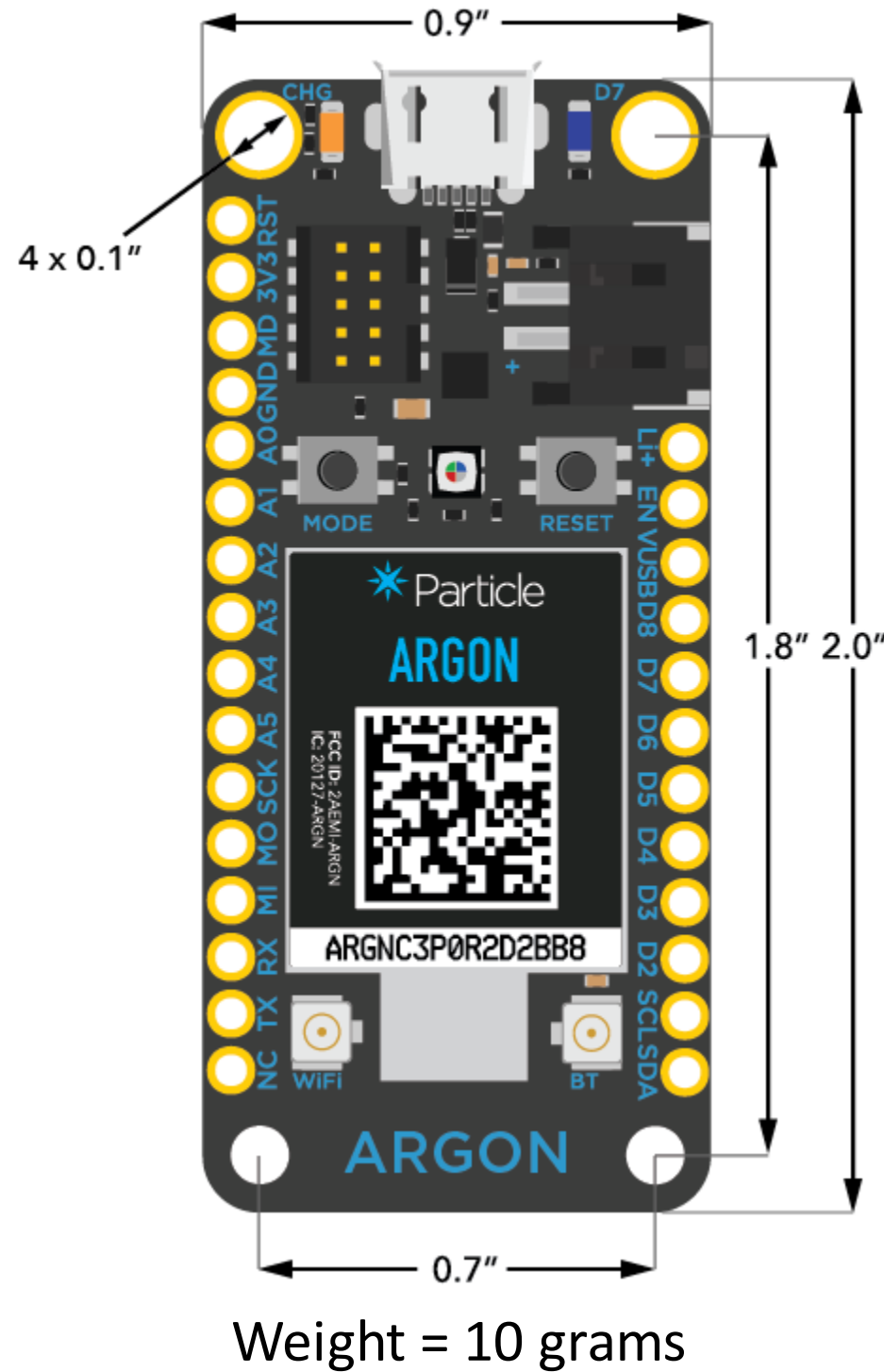
## Recommended operating conditions

| Parameter                                        | Symbol            | Min  | Typ  | Max  | Unit |
|--------------------------------------------------|-------------------|------|------|------|------|
| LiPo Battery Voltage                             | V <sub>LiPo</sub> | +3.3 |      | +4.4 | V    |
| Supply Input Voltage                             | V <sub>3V3</sub>  | +3.0 | +3.3 | +3.6 | V    |
| Supply Output Voltage                            | V <sub>3V3</sub>  |      | +3.3 |      | V    |
| Operating Temperature                            | T <sub>op</sub>   | -20  |      | +60  | °C   |
| Humidity Range Non condensing, relative humidity |                   |      |      | 95   | %    |

[1] Stresses beyond those listed under absolute maximum ratings may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under recommended operating conditions is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.



# Mechanical specifications



Standard



Qualification and approvals



# Adafruit examples of feather/wings-compliant boards

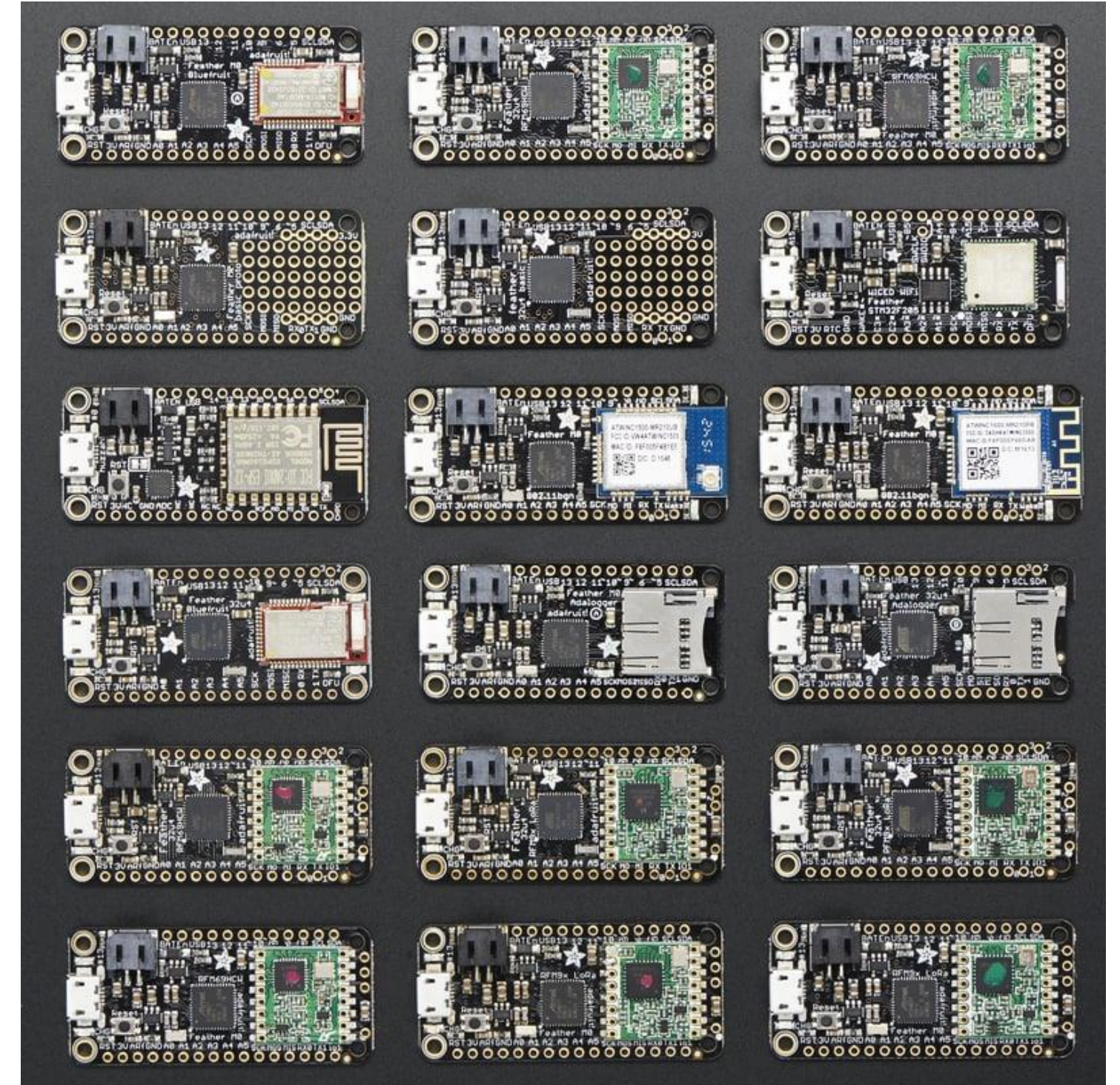
**Feather is a flexible and powerful family of microcontroller main-boards (Feathers) and daughter-boards (Wings) with a wide range of capabilities**

Main Boards - Feather(s):

<https://www.adafruit.com/category/946>

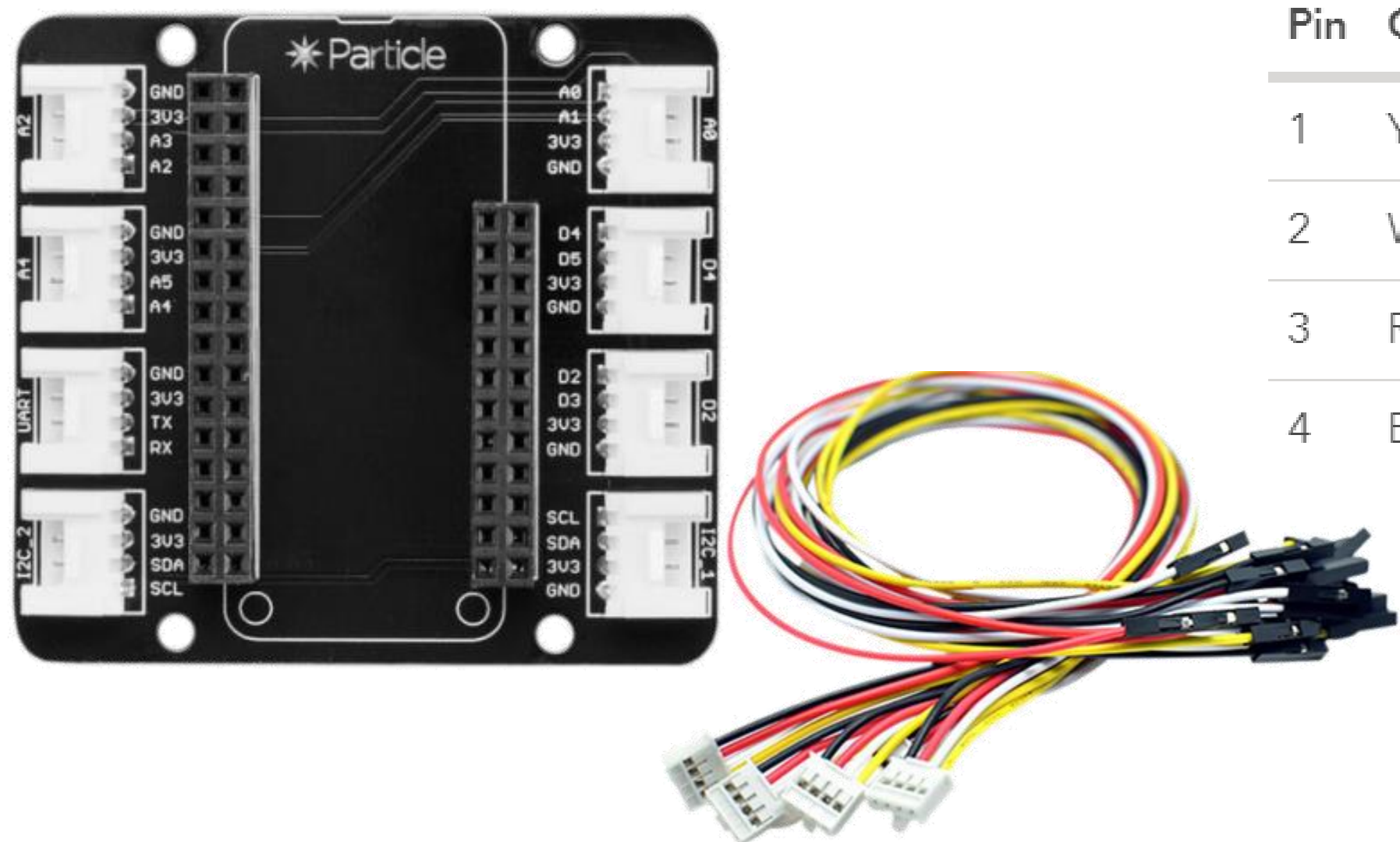
Daughter Boards FeatherWing(s):

<https://www.adafruit.com/category/945>



# Grove shield and cables

The main shield consists of a connector to plug in any of the Particle Argon or Boron dev kit and a set of 8 grove ports for plugging in various grove accessories. The ports consist of (1) 2 Digital ports, (2) 3 Analog ports, (3) 2 I2C ports, and (4) 1 UART port.



| Pin | Color  | Digital       | Analog              | I2C       | UART      |
|-----|--------|---------------|---------------------|-----------|-----------|
| 1   | Yellow | Primary I/O   | Primary Analog In   | I2C Clock | Serial RX |
| 2   | White  | Secondary I/O | Secondary Analog In | I2C Data  | Serial TX |
| 3   | Red    | Power         | Power               | Power     | Power     |
| 4   | Black  | GND           | GND                 | GND       | GND       |



# References

- Particle website: <https://www.particle.io/>
- Particle store: <https://store.particle.io/products/>
- Particle platform guides: <https://docs.particle.io/>
- Particle ARGON datasheet: <https://docs.particle.io/reference/datasheets/wi-fi/argon-datasheet/>
- Particle set-up: <https://setup.particle.io/>
- Particle «doctor»: <https://docs.particle.io/tools/doctor/>
- Particle reference manual: <https://docs.particle.io/reference/device-os/firmware/>
- Particle Web IDE: <https://build.particle.io/>
- Github repository: <https://github.com/particle-iot/argon>
- Seedstudio website: <https://www.seeedstudio.com/>





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